

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 05

Factors of Climatic Comfort

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	5.0 - Factors of Climatic Comfort (17 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Method	Question content is mapped specifically to Lecture 05, which covers subjective variables, comfort indices (ET/CET/EW/OT), Cosine Law, atmospheric depletion, humidity states/factors and air movement. Questions whose wording also demands a bioclimatic chart, detailed applications or mechanisms not present in the deck are marked Partial rather than silently completed from general knowledge.

RESULT

Lecture 05 is one of the most exam-dense compact decks. The ET-CET-EW-OT progression appears repeatedly from 2016-2023; Cosine Law and atmospheric depletion recur in 2017, 2021, 2022, 2023 and 2024; and the subjective-variable list appears in 2018 and 2024. The lecture also provides supporting humidity and air-movement concepts.

Prepared as part of the cumulative lecture-by-lecture question bank analysis.

1. Executive Summary

Lecture 05 shifts from the physiological “why” of comfort to the environmental scales and variables used to describe it. It introduces subjective variables, then develops ET, CET, EW and OT, followed by solar-incidence and atmospheric-depletion concepts. The final slides summarize humidity states/factors and global air-movement patterns. This produces several independent short-note and medium-answer question families.

17	17	9	3	5	149
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Lecture structure and exam value

Slides	Lecture block	Historical signals	Exam value
2-3	Climatic comfort + subjective variables	2016 Q6(a) partial; 2018 Q7(b); 2024 Q3(d)	Slide 3 is a direct, memorization-friendly subjective-variable list.
4-8	Comfort scales: ET, CET, EW, OT	2016 Q6(c); 2017 Q7(a-v); 2018 Q7(c); 2021 Q6(c); 2022 Q6(c); 2023 Q5(b)	Highest-return block. Learn progression and what variable each new index adds.
9-10	Cosine Law + atmospheric depletion	2017 Q7(a-i,ii); 2021 Q6(b); 2022 Q5(c), Q6(b); 2023 Q5(c); 2024 Q6(c)	Repeated theory/analytical block; diagrams improve answers.
11-13	Indirect radiation/absorption + humidity states/factors	2017 supersaturated-air short note belongs elsewhere; 2020 humidity interaction support	Useful support, but several exam questions extend beyond this deck.
14-16	Air movement causes, wind belts, effect on temperature/humidity	2020 Q1(b) partial; global-wind questions only partial	Useful conceptual support, not a substitute for the dedicated global-climate lecture.
17	Reference	No direct question	Navigation only.

Priority note

Study Slides 4-10 first. That block alone covers two of the most repeated short/medium-answer families: comfort indices and solar-radiation concepts.

2. Detailed Year-wise Question Mapping

Original paper references are preserved. Match labels mean: Exact = the slide block substantially answers the task; Direct = the requested concept is clearly taught but some expansion/sketch/application may still be needed; Partial = the deck only supports part of the requested answer.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q6(a)	15	1	What are the factors of Climatic Comfort? Explain with examples.	Partial	2-3, 11-16	The lecture emphasizes temperature, humidity and ventilation plus related modifiers; the broader four-factor formulation including radiation is clearer in Lecture 03.
2016	Q6(c)	15	1	Explain ET, CET and Bioclimatic Chart.	Partial	4-6	ET and CET are directly taught, but the bioclimatic chart construction is not provided in this deck.
2017	Q7(a-i)	5	2	Short note: The Cosine Law.	Exact	9	Slide 9 states the law and includes an incidence-angle diagram.
2017	Q7(a-ii)	5	2	Short note: Atmospheric Depletion.	Direct	10	Slide 10 gives the atmospheric-radiation balance diagram; a verbal definition may still need class explanation.
2017	Q7(a-v)	5	2	Short note: Corrected Effective Temperature.	Exact	6	Slide 6 directly defines CET as ET corrected for radiation and notes its wide use.
2018	Q7(b)	5	4	Outline the various subjective variables by which our thermal preferences are influenced.	Exact	3	Slide 3 gives the full list: clothing, acclimatization, age/sex, body shape/fat, health, food/drink, skin colour, etc.
2018	Q7(c)	10	4	What is Effective Temperature? Discuss its limitations.	Partial	5-8	Slide 5 defines ET and its three variables, but its limitations are not fully listed.
2020	Q1(b)	15	5	Describe how humidity level interacts with conduction, convection, radiation and evaporation, water body, altitude, precipitation, air movement and vegetation.	Partial	11-16	The deck covers altitude, precipitation, vegetation, water bodies and air movement as humidity/temperature modifiers, but not the full heat-transfer interaction requested.
2021	Q6(b)	10	7	Explain atmospheric depletion and Cosine Law with necessary sketches.	Exact	9-10	Both requested concepts are the central content of Slides 9-10.

2. Detailed Year-wise Question Mapping - Continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2021	Q6(c)	10	7	Differentiate between effective and corrected effective temperature and equivalent warmth.	Exact	5-7	Slides 5-7 provide the exact progression and variables.
2022	Q5(c)	5	8	Illustrate the Cosine Law and explain how the Cosine Law is used in designing buildings.	Direct	9	Slide 9 gives the law and incidence geometry; the building-design applications require brief interpretation beyond the slide.
2022	Q6(b)	10	9	Explain how atmospheric depletion and elevation affect the amount of radiation received.	Partial	10-11	Atmospheric depletion is direct; altitude is listed as a radiation/absorption factor but the mechanism is not developed.
2022	Q6(c)	10	9	Explain the progressive development of ET, CET, EW and OT and the factors integrated in different indices.	Exact	5-8	One-to-one match with the slide sequence.
2023	Q5(b)	12	11	Explain the progressive stages of development of comfort scales ET, CET, EW and OT, and how climatic factors are integrated.	Exact	5-8	Direct repeat of the index-development block.
2023	Q5(c)	5	11	Illustrate atmospheric depletion.	Exact	10	Slide 10 is an illustration-based atmospheric-depletion diagram.
2024	Q3(d)	5	13	Outline the various subjective variables by which our thermal preferences are influenced.	Exact	3	Direct repeat of the Slide 3 list.
2024	Q6(c)	7	14	Describe the Cosine Law and discuss how it can be used in built environments.	Direct	9	The law and geometry are direct; specific built-environment applications require brief design interpretation beyond the slide.

3. Repeated Question Patterns and Quick Recognition

These patterns compress historical wording into the question families that matter for revision.

P	Normalized repeated question	Years seen	Slides	Why it matters
1	Progressive development / comparison of ET, CET, EW and OT	2016, 2017, 2018, 2021, 2022, 2023	4-8	Most repeated Lecture 05 family; appears as short note, difference, progression and limitations.
2	Cosine Law and atmospheric depletion	2017, 2021, 2022, 2023, 2024	9-10	Highly repeated analytical short-note family; diagram-friendly.
3	Subjective variables influencing thermal preference	2018, 2024	3	Exact list question; very high marks-per-minute.
4	Humidity / environmental modifiers	2020 + support elsewhere	11-13	Often combined with broader climate questions; useful supporting material.
5	Air movement causes and wind pattern	2020 support; global-wind family elsewhere	14-16	Conceptual bridge to the global-climate lecture, but not a complete global-wind answer.

Quick recognition rules

Question clue	Go first to	What Lecture 05 can do
"ET / CET / EW / OT"	Slides 5-8	Exact progression. State variables added/neglected at each step.
"Corrected Effective Temperature"	Slide 6	Exact definition: ET rectified by incorporating radiation effect.
"subjective variables"	Slide 3	Exact memorization list.
"Cosine Law"	Slide 9	State intensity on tilted surface = normal intensity × cosine of incidence angle; redraw diagram.
"atmospheric depletion"	Slide 10	Use radiation-flow diagram; add one-line role of atmosphere.
"saturated / unsaturated / supersaturated"	Slide 12	Definitions are available, but historical supersaturated-air short note is better paired with humidity lecture material.
"humidity factors"	Slides 11-13	Soil, vegetation, altitude, water body, air movement, precipitation.
"global wind pattern / Coriolis"	Slides 14-15 only as support	Use dedicated global-climate lecture for full world-wind answer.

4. Slide-to-Question Reverse Index

Use this page when studying directly from the lecture deck: open the slide block, then practice the linked question family immediately.

Slides	Topic	Past questions	Study note
2	Climatic comfort: temperature, humidity, ventilation	2016 Q6(a) partial	Use as simplified intro; Lecture 03 gives the fuller comfort-factor formulation.
3	Subjective variables	2018 Q7(b); 2024 Q3(d)	One of the easiest exact-answer slides in the course.
4	Index roadmap: ET/CET/EW/OT + Cosine + depletion	Multiple 2016-2024 questions	Use as navigation slide; do not answer from it alone.
5	Effective Temperature	2016 Q6(c); 2018 Q7(c); 2021 Q6(c); 2022/2023 progression	Remember: integrates temperature + humidity + air movement.
6	Corrected Effective Temperature	2016 Q6(c); 2017 short note; 2021/2022/2023 progression	Key addition: radiation effect.
7	Equivalent Warmth	2021 Q6(c); 2022/2023 progression	Includes air temp, humidity, mean radiant temp + skin/clothing surface response; limitation about air movement at high RH is on slide.
8	Operative Temperature	2022/2023 progression	Combines radiation + air temperature when humidity/air movement are negligible.
9	Cosine Law	2017 Q7(a-i); 2021 Q6(b); 2022 Q5(c); 2024 Q6(c)	Draw the incidence-angle sketch; applications need brief design interpretation.
10	Atmospheric depletion	2017 Q7(a-ii); 2021 Q6(b); 2022 Q6(b); 2023 Q5(c)	High-return illustration slide.
11-13	Radiation absorption + humidity states/factors	2020 Q1(b) partial	Good supporting detail for longer climate-interaction answers.
14-16	Air movement: causes, wind belts, environmental effects	2020 Q1(b) partial; global-wind support	Bridge material; do not replace dedicated global-climate lecture.
17	Reference	No direct question	No study time needed.

5. Coverage Gaps and Diagnostic Exclusions

Lecture 05 contains several famous terms, but not every historical question using those terms is fully answerable from this deck. The main gaps are chart construction, detailed limitations and broader global-wind mechanisms.

Gap	Affected questions	Action
Bioclimatic chart construction/interpretation	2016 Q6(c)	ET/CET are here; the chart itself is not taught in this deck. Use class notes/reference material if required.
Full limitations of Effective Temperature	2018 Q7(c)	Slide 5 defines ET but does not provide a systematic limitations list.
Built-environment applications of Cosine Law	2022 Q5(c); 2024 Q6(c)	The law is direct; add shading/orientation/solar-incidence applications from later design lectures.
Elevation mechanism within atmospheric depletion/radiation	2022 Q6(b)	Altitude is only listed as a factor on Slide 11; detailed mechanism is not developed.
Full heat-transfer interaction in humidity question	2020 Q1(b)	Use Lecture 03 for conduction/convection/radiation/evaporation mechanisms.
Complete global wind / Coriolis explanation	Any world-wind long answer	Slides 14-15 are only a summary; use the dedicated Global Climatic Factors lecture.

Mapping discipline

The deck is treated as authoritative for the variables and sequence it explicitly states. Where the paper asks for a chart, detailed limitation or design application not printed in the deck, the mapping is downgraded to Partial/Direct rather than silently filled from outside knowledge.

6. Recommended Study Plan

Study this lecture in two high-yield clusters: first comfort indices, then Cosine Law/atmospheric depletion. Subjective variables are a quick exact-answer bonus.

Step	Study block	Slides	Practice immediately
1	ET -> CET -> EW -> OT progression	5-8	2016 Q6(c); 2017 Q7(a-v); 2018 Q7(c); 2021 Q6(c); 2022 Q6(c); 2023 Q5(b)
2	Cosine Law	9	2017 Q7(a-i); 2021 Q6(b); 2022 Q5(c); 2024 Q6(c)
3	Atmospheric depletion	10	2017 Q7(a-ii); 2021 Q6(b); 2022 Q6(b); 2023 Q5(c)
4	Subjective variables	3	2018 Q7(b); 2024 Q3(d)
5	Humidity states/factors + air movement summary	11-16	2020 Q1(b) support; use as bridge to other lectures

Exam-ready minimum from this lecture

Priority	What to reproduce	Exam technique
A	ET/CET/EW/OT comparison	Use a four-row table: variables included, improvement over previous scale, key limitation.
A	Cosine Law sketch	Write the relationship in words/equation and show incidence angle.
A	Atmospheric depletion diagram	Label incoming, reflected/absorbed, surface receipt and outgoing components.
B	Subjective variables	Memorize the seven-item list from Slide 3.

Bottom line

Lecture 05 is a scoring lecture because the same terms repeat with only small wording changes. Master Slides 3 and 5-10 and you cover most of its historical exam value.